PASSES: A Unified, GPU-Accelerated Spectral Framework for Coupled Aeroelastic, Thermodynamic, and Stochastic GNC Flight Simulation

Ethan Knox*[1]

[1]Independent Researcher, Champaign, IL, USA

August 1, 2026

Abstract

The design and validation of atmospheric aerospace vehicles requires the simultaneous evaluation of tightly coupled physical domains: structural aeroelasticity, thermal ablation, and active Guidance, Navigation, and Control (GNC). Simulation frameworks that treat these domains with separate solvers and exchange data through mesh interpolation incur spatial interpolation error at every coupling interface, and frameworks that track receding ablation fronts on a deforming mesh incur remeshing cost that scales poorly under Monte Carlo replication. This paper presents PASSES, a flight simulation framework in which every spatial domain is discretized once, on a fixed computational grid, and held fixed for the duration of the trajectory.

Three formulations make this possible. Structural bending under spatially varying flexural rigidity is discretized by Chebyshev spectral collocation, with free-free boundary conditions imposed by projection onto the null space of the constraint operator rather than by row replacement. Thermal ablation is governed by a two-phase charring model in the style of the CMA framework, in which a Landau coordinate transformation maps the receding physical domain onto a fixed computational domain, so that surface recession appears as a grid-velocity term rather than as a moving boundary. Localized slosh forces are regularized by a quadrature-normalized Gaussian kernel that transfers force exactly under the collocation quadrature rule. The resulting semi-discrete system is a single system of ordinary differential equations in one global state vector.

For navigation resilience through stage separation, the GNC loop pairs a χ^2 anomaly test on the squared Mahalanobis distance of the innovation with Innovation-Based Adaptive Estimation, using a bounded trace-ratio inflation factor. Terminal guidance uses Aerodynamically-Compensated Augmented Proportional Navigation with a numerically stable time-to-go root and feed-forward gravity compensation. We give the stiffness bound that governs admissible explicit time steps for the spectral operator, and specify the estimator used to extract Circular Error Probable and 95% containment radii from Monte Carlo terminal dispersions, including its validity range and sampling error.

This manuscript presents the formulation and the verification plan. Numerical verification, validation, and performance results are identified explicitly throughout and are the subject of the companion experimental report.

Keywords: Spectral Collocation, Multi-Physics Simulation, Method of Lines, Innovation-Based Adaptive Estimation, Extended Kalman Filter, Proportional Navigation, Monte Carlo Covariance Estimation, Applied Mathematics.

AMS Subject Classification: 65M70, 93E11, 74F10, 80A22.

*Email: ethank5149@gmail.com

Contents

Nomenclature

Symbol	Definition
Structural	
$w(x,t)$	transverse deflection of the beam axis, m
$EI(x)$	effective flexural rigidity, $\mathrm{N\,m^2}$
$m(x)$	mass per unit length, kg/m
$q(x,t)$	distributed transverse load, N/m
L	vehicle reference length, m
ξ	computational coordinate, $\xi \in [-1,1]$
N	spectral order (number of collocation intervals)
$\mathbf{D}$	Chebyshev differentiation matrix on ξ, $(N+1) \times (N+1)$
$\mathbf{K}, \mathbf{M}$	assembled stiffness and mass matrices
$\mathbf{B}$	boundary constraint matrix
$\mathbf{Z}$	orthonormal basis for $\ker \mathbf{B}$
$\omega_{\max}$	largest structural angular frequency, rad/s
$\kappa(\cdot)$	spectral condition number
Slosh	
$x_s^{(k)}$	axial station of the k-th slosh mass, m
$F_{\text{slosh}}^{(k)}$	lateral force from the k-th slosh mass, N
δ_σ	Gaussian regularized Dirac kernel, $\mathrm{m^{-1}}$
σ	kernel bandwidth, m
w_j^{CC}	Clenshaw–Curtis quadrature weight at node j
Thermal	
y	depth coordinate measured from the original surface, m
$s(t)$	surface recession depth, m
η	Landau-transformed depth coordinate, $\eta \in [0,1]$
T	temperature, K
ρ_v, ρ_c	virgin and char densities, $\mathrm{kg/m^3}$
$\dot{m}_g$	pyrolysis gas mass flux, $\mathrm{kg/(m^2\,s)}$
$h_g, \bar{h}$	pyrolysis gas and solid enthalpies, J/kg
B'	non-dimensional mass blowing rate
ϕ	blowing reduction factor
$\mathbf{H}$	vector of nodal in-depth enthalpies (thermal state)
Rigid body and GNC	
$\mathbf{r}_E, \mathbf{v}_E$	ECEF position and velocity, m, m/s
$\mathbf{q}_{E2B}$	attitude quaternion, Earth to body
$\boldsymbol{\omega}_B$	body angular rate, rad/s
m_{bulk}	rigid-body mass, kg
$\boldsymbol{\nu}_k$	measurement innovation
$\mathbf{S}_k$	theoretical innovation covariance
$\hat{\mathbf{C}}_k$	sample innovation covariance over window N_w
d_k^2	squared Mahalanobis distance of the innovation
α_k	process noise inflation factor
$\mathbf{Q}, \mathbf{R}$	process and measurement noise covariances
V_c, A_c	closing velocity and closing acceleration, m/s, $\mathrm{m/s^2}$
R_{LOS}	line-of-sight range, m
t_{go}	time-to-go, s

Symbol	Definition
Statistics	
$\mathbf{\Sigma}$	terminal impact covariance, m^2
λ_1, λ_2	eigenvalues of $\mathbf{\Sigma}$, $\lambda_1 \geq \lambda_2$
R_{95}	95% containment radius, m
N_{MC}	Monte Carlo sample count

1 Introduction

The design and validation of atmospheric aerospace vehicles requires the simultaneous, high-fidelity evaluation of tightly coupled physical domains. During atmospheric maneuvering and terminal descent, a vehicle is subjected to severe aerodynamic loading, thermodynamic heating, and stochastic environmental disturbance. Historically, flight simulation architectures have compartmentalized these domains, treating structural aeroelasticity, thermal ablation, and Guidance, Navigation, and Control (GNC) as separate solvers exchanging boundary data. This partitioning is effective for steady-state analysis, but it introduces two costs that become dominant in the regimes of interest here.

The first cost is interpolation error at coupling interfaces. When structural, thermal, and aerodynamic domains are discretized on distinct meshes, every exchange of surface pressure, heat flux, or displacement requires spatial interpolation, and the resulting field is generally no smoother than C^0 across element boundaries. Gradients taken of an interpolated field — and the aerodynamic and thermal closures require several — amplify this error.

The second cost is remeshing. Ablative recession and large structural deformation are conventionally handled by moving-boundary formulations with periodic mesh regeneration [22, 5]. Remeshing is expensive, it is difficult to vectorize, and it introduces a projection step between the pre- and post-remesh solution that is a further source of non-smoothness. For a single deterministic trajectory this is tolerable. For the 10^3–10^4 replicate trajectories required to resolve a terminal dispersion footprint, it is not.

This paper presents PASSES (Parameterized AeroStructural State-Estimation Sandbox), a framework in which every spatial domain is discretized once, on a fixed computational grid, and held fixed for the entire trajectory. The framework is built on three formulations that make a fixed grid possible without sacrificing the physics:

1. **Structural.** Bending under spatially varying flexural rigidity is discretized by Chebyshev spectral collocation [25], with free-free boundary conditions imposed by projection onto the null space of the boundary constraint operator [11] rather than by row replacement. Projection preserves the symmetry of the reduced operator, which row replacement destroys.

2. **Thermal.** Ablation is governed by a two-phase charring model in the style of the CMA framework [19, 4]. A Landau coordinate transformation maps the receding physical domain onto a fixed computational domain, so surface recession enters the energy equation as a grid-velocity advection term rather than as a moving boundary. This is what eliminates front tracking; the front still exists physically, but it is stationary in computational coordinates.

3. **Coupling.** Localized slosh forces, which are point loads on a continuous beam, are regularized by a Gaussian kernel normalized against the collocation quadrature rule, so that total force is transferred exactly rather than to within quadrature error.

The resulting semi-discrete system is a single system of ordinary differential equations in one global state vector, integrated by the Method of Lines. Section 3.6 addresses a consequence of this choice that is frequently left implicit in the spectral literature: the fourth-order collocation operator has an eigenvalue spread that imposes a severe explicit step-size limit, and we state that limit rather than assume it away.

A physics formulation is of limited value if the GNC logic built on it cannot survive the transients the physics produces. During stage separation, instantaneous mass loss and structural ringing drive measurement innovations far outside the covariance predicted by the filter, and a fixed-gain Extended Kalman Filter (EKF) will diverge. PASSES pairs a χ^2 test on the squared Mahalanobis distance of the innovation [2] with Innovation-Based Adaptive Estimation [15, 17, 10], inflating process noise by a bounded trace ratio while the anomaly persists. Terminal guidance uses Aerodynamically-Compensated Augmented Proportional Navigation with a numerically stable time-to-go root and feed-forward gravity compensation [26].

1.1 Scope of this manuscript

This manuscript presents the mathematical formulation, the discretization, the stability constraints that follow from it, and the verification and validation plan. Since the plan was written, a reference implementation has been built and the verification tasks executed. Where a claim here has been measured, the measured value is quoted inline and identified as such; where implementation revealed a claim to be imprecise or wrong, it has been corrected in place and the correction flagged rather than silently absorbed (see Remark 3.9 and the conditioning discussion in Remark 3.3). Convergence rates, condition numbers, filter recovery times, dispersion footprints and throughput figures quoted below are measurements; detailed protocols, figures and tabulations are the subject of a companion experimental report. Claims presented without a measurement remain restricted to what follows analytically or by direct reference to prior literature.

A companion manuscript [13] extends the same fixed-grid principle to six-degree-of-freedom hypersonic glide and fractional orbital trajectories, replacing the Euler–Bernoulli beam with a Mindlin–Reissner anisotropic plate and the dense Chebyshev operator with a sparse ultraspherical one.

2 Related Work

The individual components of PASSES are established. The contribution claimed here is their combination under a single fixed-grid, single-state-vector formulation, and the consequences that follow for stochastic batch execution.

Ablation and thermal response. The CMA framework of Moyer and Rindal [19] established the two-phase charring formulation with in-depth pyrolysis and surface energy balance that remains the basis of production tools. FIAT [4] provides a fully implicit one-dimensional implementation. Dec and Braun [6] give an approximate sizing formulation for entry system design. These tools solve the thermal problem to a fidelity PASSES does not attempt to exceed; the difference is that they are one-dimensional stand-alone solvers coupled to trajectory codes by file exchange, whereas here the thermal state is a block of the same ODE state vector as the structural and rigid-body states and is advanced by the same integrator.

Spectral methods for structural operators. Trefethen [25] is the standard reference for Chebyshev collocation. The principal difficulty for fourth-order operators is conditioning: the dense collocation representation of $\mathrm{d}^4/\mathrm{d}x^4$ has a condition number growing as $\mathcal{O}(N^8)$. Two families of remedy exist. Driscoll and Hale [8] resample onto a rectangular grid so that boundary conditions are appended rather than substituted. Olver and Townsend [20] reformulate in an ultraspherical basis in which the differential operator is banded and the conditioning is $\mathcal{O}(1)$. PASSES uses the null-space projection of Hsu et al. [11], which retains the dense collocation basis but preserves operator symmetry; the companion HGV paper adopts the ultraspherical route instead, for reasons of sparsity in the bivariate case.

Adaptive filtering. Mehra [15] established the identification of unknown process and measurement covariances from the innovation sequence, and surveyed the resulting family of approaches in [14]. Mohamed and Schwarz [17] developed the innovation-based variant for INS/GPS integration, and Hide et al. [10] characterized its behavior with low-cost inertial sensors. The adaptation used here is a deliberately conservative special case: rather than estimating $\mathbf{Q}$ entrywise, which is prone to indefiniteness, it estimates a single scalar inflation factor.

Trajectory optimization. Pseudospectral discretization of optimal control [9, 3, 12] shares the spectral machinery used here for the structural operator. PASSES does not currently solve an optimal control problem in the loop; the companion HGV paper does.

Positioning. The closest production analogue to the combination presented here is a coupled FIAT/Nastran/POST-class toolchain. Such a toolchain is more mature and more extensively validated in each individual domain. The claim made for PASSES is narrower and specific: because the coupled system is a single ODE with a fixed sparsity pattern and no mid-flight remeshing or matrix refactorization, a replicate batch is a batched tensor operation, which is what makes the Monte Carlo sample sizes in section 6 tractable on a single accelerator. Whether that advantage survives contact with the implementation is the question the companion experimental report exists to answer.

3 Mathematical Foundations: The Unified Physics Engine

3.1 Aeroelastic dynamics and spectral collocation

The structural bending dynamics are governed by an Euler–Bernoulli beam with spatially varying mass and flexural rigidity. Let $w(x,t)$ denote transverse deflection over the physical domain $x \in [0, L]$. The governing partial differential equation is

$$\frac{\partial^2}{\partial x^2}\left(EI(x)\frac{\partial^2 w(x,t)}{\partial x^2}\right) + m(x)\frac{\partial^2 w(x,t)}{\partial t^2} = q(x,t). \tag{3.1}$$

Assumption 3.1. Deflections are small relative to L, so that the linear curvature-displacement relation holds; cross-sections remain plane and normal to the deformed axis, so transverse shear and rotary inertia are neglected; and $EI(x) \in C^2([0, L])$ with $EI(x) > 0$. The last condition is what permits the product-rule expansion below and is the reason the material property fields are constructed by hyperbolic blending rather than by piecewise assignment.

Assumption 3.1 is a genuine restriction. Euler–Bernoulli theory loses accuracy when the section depth is not small relative to the bending wavelength, which is exactly the regime of the thick hull sections addressed in the companion paper by a Mindlin–Reissner formulation.

The physical domain is mapped to the Chebyshev computational domain $\xi \in [-1, 1]$ by the affine transformation

$$x = \frac{L}{2}(\xi + 1), \qquad \frac{\mathrm{d}}{\mathrm{d}x} = \frac{2}{L}\frac{\mathrm{d}}{\mathrm{d}\xi}, \tag{3.2}$$

so that a differentiation matrix $\mathbf{D}$ constructed on ξ represents $\mathrm{d}/\mathrm{d}x$ as $(2/L)\mathbf{D}$. All operators below are written in computational coordinates with the scaling factor absorbed; the factor $(2/L)^4$ on the fourth-derivative term must be carried explicitly in implementation and is a common source of error.

Expanding the outer derivative in eq. (3.1) by the product rule,

$$\frac{\partial^2}{\partial x^2}\left(EI\frac{\partial^2 w}{\partial x^2}\right) = EI\frac{\partial^4 w}{\partial x^4} + 2\frac{\partial EI}{\partial x}\frac{\partial^3 w}{\partial x^3} + \frac{\partial^2 EI}{\partial x^2}\frac{\partial^2 w}{\partial x^2}. \tag{3.3}$$

Let $\mathbf{EI} \in \mathbb{R}^{N+1}$ denote the vector of rigidity values sampled at the Chebyshev–Gauss–Lobatto nodes $\xi_j = \cos(j\pi/N)$, $j = 0, \dots, N$. Collocating eq. (3.3) at those nodes gives the assembled stiffness operator

$$\mathbf{K} = \mathrm{diag}(\mathbf{EI})\,\mathbf{D}^4 + 2\,\mathrm{diag}(\mathbf{D}\,\mathbf{EI})\,\mathbf{D}^3 + \mathrm{diag}(\mathbf{D}^2\mathbf{EI})\,\mathbf{D}^2, \tag{3.4}$$

with mass operator $\mathbf{M} = \operatorname{diag}(\mathbf{m})$. Retaining all three terms of eq. (3.3), rather than the leading term alone, is necessary whenever EI varies appreciably over a bending wavelength, which it does across stage joints and in regions of localized thermal softening.

Remark 3.2. Equation (3.4) is not symmetric, because $\operatorname{diag}(\mathbf{EI})\mathbf{D}^4$ is not. The self-adjointness of the continuous operator is recovered in the weak form; the collocation form used here trades that structure for simplicity of assembly. The null-space projection of section 3.2 preserves whatever symmetry the unprojected operator possesses but does not create symmetry that was absent.

3.2 Free-free boundary conditions by null-space projection

An unconstrained vehicle in flight satisfies free-free boundary conditions: zero bending moment and zero shear at both ends,

$$EI\frac{\partial^2 w}{\partial x^2}\bigg|_{x=0,L} = 0, \qquad \frac{\partial}{\partial x}\left(EI\frac{\partial^2 w}{\partial x^2}\right)\bigg|_{x=0,L} = 0. \tag{3.5}$$

Collocating eq. (3.5) yields four linear constraints, written $\mathbf{B}\mathbf{w} = \mathbf{0}$ with $\mathbf{B} \in \mathbb{R}^{4\times(N+1)}$.

The conventional treatment replaces four rows of $\mathbf{K}$ with the rows of $\mathbf{B}$. This enforces the constraints but destroys any symmetry of $\mathbf{K}$ and produces complex eigenvalues with positive real parts for the free-free case, which manifest as spurious growth in time integration. The null-space approach [11] instead computes an orthonormal basis $\mathbf{Z} \in \mathbb{R}^{(N+1)\times(N-3)}$ for $\ker \mathbf{B}$ — in practice from the trailing right singular vectors of $\mathbf{B}$, or from a rank-revealing QR factorization of $\mathbf{B}^\top$ — and substitutes $\mathbf{w} = \mathbf{Z}\hat{\mathbf{w}}$. Every admissible $\hat{\mathbf{w}}$ then satisfies the boundary conditions identically, since $\mathbf{B}\mathbf{Z} = \mathbf{0}$ by construction. Left-multiplying by $\mathbf{Z}^\top$ gives the reduced system

$$\underbrace{\mathbf{Z}^\top\mathbf{M}\mathbf{Z}}_{\hat{\mathbf{M}}}\,\ddot{\hat{\mathbf{w}}} + \underbrace{\mathbf{Z}^\top\mathbf{K}\mathbf{Z}}_{\hat{\mathbf{K}}}\,\hat{\mathbf{w}} = \mathbf{Z}^\top\mathbf{f}, \tag{3.6}$$

of dimension $N-3$. Because $\mathbf{Z}$ has orthonormal columns, $\hat{\mathbf{M}} \succ 0$ whenever $\mathbf{M} \succ 0$, and the reduced problem is a well-posed generalized eigenvalue problem.

Remark 3.3 (Conditioning)**.** The dense Chebyshev representation of $\mathrm{d}^4/\mathrm{d}x^4$ has $\kappa(\mathbf{D}^4) = \mathcal{O}(N^8)$. Projection onto $\ker\mathbf{B}$ removes the four modes most responsible for the extremal singular values, and Hsu et al. [11] report substantially improved conditioning in the beam problems they examine. We deliberately do not assert a sharp asymptotic rate for $\kappa(\hat{\mathbf{K}})$ here: we have not proved one, and the constant depends on the rigidity profile $EI(x)$. Measuring $\kappa(\hat{\mathbf{K}})$ against N for the representative profiles of this study is verification task V1 in section 7.

Measured, the answer has two parts, and the distinction turns out to matter. The raw $\kappa_2(\hat{\mathbf{K}})$ is pinned at the reciprocal rounding floor, $\sim 1/\varepsilon$, at *every* N: the free-free operator retains its two *physical* rigid-body null directions, so its smallest singular value is rounding noise by construction and the raw condition number carries no information about the discretization. The informative quantity is the condition number of the regular part, σ_1/σ_{n-2}, and that grows as $N^{8.0}$ for both the uniform and the stepped profiles. The projection removes the constraint-violating extremal modes — and the free-free frequencies come within 1.2×10^{-9} of the analytic values at $N = 32$, against a 10^{-6} criterion — but it does not flatten the asymptotic rate. Reporting a raw condition number for an operator with a physical null space measures the floating-point format, not the method.

3.3 Fluid slosh and quadrature-consistent force regularization

Liquid propellant motion is modeled by equivalent spring-mass-damper elements at axial stations $x_s^{(k)}$, $k = 1,\ldots,N_{\text{tank}}$, following the standard mechanical-analogue treatment [1, 7, 21]. Each element applies a lateral force $F_{\text{slosh}}^{(k)}(t)$ to the structure at a point.

A point load on a spectral grid is a Dirac delta, whose spectral expansion has coefficients that do not decay, producing Gibbs oscillations across the entire domain. The load is therefore regularized by a Gaussian kernel,

$$\delta_\sigma(x - x_s) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x - x_s)^2}{2\sigma^2}\right), \tag{3.7}$$

giving the distributed load

$$q_{\text{slosh}}(x, t) = \sum_{k=1}^{N_{\text{tank}}} F_{\text{slosh}}^{(k)}(t)\, \tilde{\delta}_\sigma\left(x - x_s^{(k)}\right). \tag{3.8}$$

Two details determine whether this is admissible, and both are frequently omitted.

Bandwidth. Chebyshev–Gauss–Lobatto nodes are not uniformly spaced: local spacing is $h(x) \approx \pi\sqrt{1 - \xi^2}/N$, which is $\mathcal{O}(N^{-1})$ near the center and $\mathcal{O}(N^{-2})$ near the ends. A single global σ is therefore either under-resolved at the center or wastefully wide at the ends. We set $\sigma^{(k)} = \gamma\, h(x_s^{(k)})$ with $\gamma \in [1, 2]$, so the kernel is resolved by at least a few nodes wherever the tank happens to sit. Taking γ below 1 reintroduces the oscillations the kernel was meant to suppress; taking it large over-smears the load and artificially stiffens the response.

Exact force transfer. The Gaussian in eq. (3.7) integrates to unity over $\mathbb{R}$, but not over the truncated domain $[0, L]$, and in any case the discrete system sees not the integral but the quadrature sum. Applying eq. (3.7) directly therefore transfers slightly less than $F_{\text{slosh}}^{(k)}$ to the structure, and the deficit varies with $x_s^{(k)}$ as the tank drains — a spurious, slowly varying force error. We instead use the discretely normalized kernel

$$\tilde{\delta}_\sigma(x_j - x_s) = \frac{\delta_\sigma(x_j - x_s)}{\sum_{i=0}^{N} w_i^{\text{CC}}\, \delta_\sigma(x_i - x_s)}, \tag{3.9}$$

where w_i^{CC} are the Clenshaw–Curtis weights associated with the collocation nodes.

Proposition 3.4 (Exact force transfer). *With $\tilde{\delta}_\sigma$ defined by eq.* (3.9)*, the total transverse force delivered to the discrete structural system equals $\sum_k F_{\text{slosh}}^{(k)}$ exactly, independent of σ, N, and the station $x_s^{(k)}$.*

Proof. The discrete total force is $\sum_j w_j^{\text{CC}} q_{\text{slosh}}(x_j)$. Substituting eq. (3.8) and eq. (3.9) and exchanging the order of summation, the inner sum $\sum_j w_j^{\text{CC}} \tilde{\delta}_\sigma(x_j - x_s^{(k)})$ equals unity by construction of the denominator, leaving $\sum_k F_{\text{slosh}}^{(k)}$. □

Remark 3.5. Proposition 3.4 guarantees the zeroth moment only. The first moment — the applied bending moment about the vehicle center of mass — is transferred to within the quadrature error of the kernel's first moment, which is $\mathcal{O}(\sigma^2)$ for a symmetric kernel in the domain interior but degrades near the endpoints where the kernel is truncated asymmetrically. Tanks located within $\sim 2\sigma$ of a vehicle end should be expected to exhibit a small moment bias. Quantifying it is verification task V2 in section 7.

Measured, the interior behavior is better than the bound above suggests. For a kernel resolved by the grid, Clenshaw–Curtis integrates the Gaussian's first moment spectrally, so the interior moment error sits near the rounding floor rather than at $\mathcal{O}(\sigma^2)$; the $\mathcal{O}(\sigma^2)$ allowance is consumed only where truncation destroys the kernel's symmetry. The endpoint bias is real and behaves as predicted: at $x_s/\sigma = 2$ it is 2.2×10^{-3} relative and grows monotonically as the station approaches the end. Total force transfer is exact everywhere, worst case 1.5×10^{-16} relative across N, γ, and station.

3.4 Charring ablation on a fixed computational domain

Composite thermal protection materials such as PICA and carbon-phenolic do not melt. They pyrolyze: an in-depth reaction zone of finite thickness decomposes the virgin resin, the resulting gas percolates outward through the porous char, and the char surface recedes by oxidation and mechanical removal. Modeling this as a single-temperature Stefan phase change misrepresents both the in-depth temperature profile and the surface energy balance, because it omits the pyrolysis gas enthalpy and the blowing reduction of convective heating. PASSES therefore uses a two-phase charring formulation in the style of CMA [19, 4].

3.4.1 Decomposition kinetics

The solid is treated as three components — two resin constituents (A, B) and a filler (C) — each decomposing by an independent Arrhenius law,

$$\left.\frac{\partial \rho_i}{\partial t}\right|_y = -A_i\,\rho_{v,i}\left(\frac{\rho_i-\rho_{c,i}}{\rho_{v,i}}\right)^{n_i}\exp\!\left(-\frac{E_i}{\mathcal{R}T}\right), \qquad i\in\{A,B,C\}, \tag{3.10}$$

with the bulk density recovered as $\rho = \Gamma(\rho_A+\rho_B)+(1-\Gamma)\rho_C$ for resin volume fraction Γ. The instantaneous degree of char is $\beta = (\rho_v-\rho)/(\rho_v-\rho_c)$, and the thermal conductivity, specific heat, and emissivity are interpolated between virgin and char values by β.

Remark 3.6 (Two solvers are not two independent checks)**.** Because the FIAT program itself is export-controlled, the ablation solver of this section is accompanied by a second implementation written directly from FIAT's published governing equations [4, 16]: conservative finite volume on a multilayer ply stack, backward Euler in time, and a Newton solve whose Jacobian agrees with a central difference of its own residual to that difference's truncation error. Where this paper's spectral solver and that one are run on the same problem they agree, and the finite-volume path additionally conserves energy exactly on a sealed stack, reproduces the exact series-resistance profile across a ten-to-one conductivity jump at a ply interface, and returns terminal recession converging at observed order 2.85 in cells and 0.96 in time steps — the latter being backward Euler's formal order, which is the answer one wants to see.

It is worth being exact about what this does and does not establish. Two discretizations agreeing rules out a large class of discretization errors, and it rules out essentially none of the modelling ones: both solve the same equations, with the same closures, written by the same hand. The verification plan of section 7 states V4's criterion against a *FIAT reference case*, and that criterion is untouched by this. What has changed is only that a published case, when one is obtained, can now be run through a solver of the same formulation rather than through a spectral solver of the same physics.

Three things the construction taught are not in the source papers. First, the archived scan of the blowing correction, [4] Eq. (11), drops its logarithm — the extracted form neither tends to unity as $B' \to 0$ nor reduces to the classical laminar result at $\lambda = 1/2$, both of which the surrounding text requires; [16] prints it intact. Second, the two in-depth radiation treatments the source offers as alternatives, its integrodifferential Eq. (2) and its optically thick Eq. (3), agree only once the *cell* optical depth is resolved: at $\Delta\kappa = 5$ per cell they differ by a factor of 3.8, at $\Delta\kappa = 0.5$ by six percent, and at $\Delta\kappa = 0.05$ by under one. That is a statement about discretizing Eq. (2), not a disagreement between the two models, and it is easy to mistake for one. Third, and least expected: the thermal conductivity of a porous ablator depends on the pressure of the gas in its pores, and the two published models for PICA disagree about the sign of that dependence. The MEDLI2 material-response paper [18] tabulates room-temperature conductivity at both 1 and 10^{-3} atm; the heritage model has virgin conductivity *rising* threefold as pressure falls, the re-measured model has it falling by a quarter, and at 10^{-3} atm the two differ by a factor of 4.1 — in the regime that governs entry. A property model that is a function of temperature and char fraction alone — which is what section 3.4 specifies above, and what we had implemented — can represent neither.

A fourth point is more serious, because it concerns the model form rather than its coefficients. **??** follows FIAT in writing pyrolysis as a set of *independent parallel* Arrhenius reactions. Every term in such a sum shifts its decomposition peak to higher temperature as the heating rate rises, so the sum can do nothing else. Carbon/phenolic is measured to do the opposite: above roughly 300 K/min the pyrolysis peak moves *down* in temperature, and Torres-Herrador and co-workers state that parallel mechanisms are "not able to reproduce this effect due to their mathematical formulation" [24]. Running their two published PICA models — a parallel set [23] and a competitive one [24] — at the heating rates they were calibrated against reproduces exactly that split: +105 K for the parallel form against −129 K for the competitive one. This is not a calibration error and no refitting removes it. It matters for entry rather than for the laboratory, since heating rates across an entry heat shield span 60 to 60 000 K/min while the thermogravimetric data such models are calibrated on rarely exceeds tens of K/min. We record it as a limitation of eq. (3.10) as written.

3.4.2 Landau transformation to a fixed grid

Let $s(t)$ be the recession depth. The physical thermal domain $y \in [s(t), L_{\text{TPS}}]$ shrinks as the surface recedes. Define

$$\eta = \frac{y - s(t)}{L_{\text{TPS}} - s(t)} \in [0, 1], \tag{3.11}$$

which maps the receding domain onto a fixed computational domain for all t with $s(t) < L_{\text{TPS}}$. Under eq. (3.11) the time derivative at fixed η acquires a grid-velocity term,

$$\left.\frac{\partial}{\partial t}\right|_y = \left.\frac{\partial}{\partial t}\right|_\eta - \frac{\dot{s}(1-\eta)}{L_{\text{TPS}} - s}\frac{\partial}{\partial \eta}. \tag{3.12}$$

This is the precise sense in which PASSES eliminates front tracking. The recession front is not removed from the physics; it is rendered stationary in computational coordinates, so no node is ever created, destroyed, or interpolated. The cost is the advection term in eq. (3.12), which is explicit, local, and cheap.

3.4.3 In-depth energy equation

Combining conduction, pyrolysis gas convection, decomposition enthalpy, and the grid velocity, the in-depth energy equation in computational coordinates is

$$\rho c_p \left.\frac{\partial T}{\partial t}\right|_\eta = \frac{1}{\ell^2}\frac{\partial}{\partial \eta}\left(k\frac{\partial T}{\partial \eta}\right) + \frac{1}{\ell}\left[\dot{m}_g c_{p_g} + \rho c_p \dot{s}(1-\eta)\right]\frac{\partial T}{\partial \eta} + \left(h_g - \bar{h}\right)\left.\frac{\partial \rho}{\partial t}\right|_\eta, \tag{3.13}$$

where $\ell = L_{\text{TPS}} - s(t)$. The pyrolysis gas flux follows from mass continuity in the same coordinates,

$$\frac{1}{\ell}\frac{\partial \dot{m}_g}{\partial \eta} = -\left.\frac{\partial \rho}{\partial t}\right|_\eta, \tag{3.14}$$

integrated from the back face outward with $\dot{m}_g = 0$ at $\eta = 1$.

Remark 3.7 (Which density rate sources these equations)**.** Equation (3.13) and eq. (3.14) as written source both the pyrolysis-gas continuity and the decomposition enthalpy with $\partial\rho/\partial t|_\eta$ — the full computational-frame rate, which includes the grid-advection contribution of eq. (3.12). The CMA lineage instead sources them with the material-frame Arrhenius rate alone. The two differ at $\mathcal{O}(\dot{s}\,\rho_\eta/\ell)$, which is small for slow recession but is not zero, and the distinction is invisible unless one writes the terms out. We state the convention explicitly rather than leave it to the reader to infer: the equations above are in the computational frame. An implementation should expose the choice rather than bury it.

3.4.4 Surface energy balance and blowing reduction

At $\eta = 0$ the surface balance equates net incoming flux to conduction into the solid:

$$\underbrace{\rho_e u_e C_H (h_r - h_w)\,\phi}_{\text{convective, blowing-reduced}} + \underbrace{\alpha_w \dot{q}_{\text{rad,in}}}_{\text{absorbed radiation}} + \underbrace{\dot{m}_c h_c + \dot{m}_g h_g - (\dot{m}_c + \dot{m}_g) h_w}_{\text{mass-transfer enthalpy}} - \underbrace{\varepsilon \sigma_S T_w^4}_{\text{re-radiation}} = \frac{k}{\ell} \frac{\partial T}{\partial \eta}\bigg|_{\eta=0}, \tag{3.15}$$

with char mass flux $\dot{m}_c = \rho_c \dot{s}$. The injection of pyrolysis gas and char into the boundary layer thickens it and reduces convective heating. This is captured by the standard blowing correction

$$\phi = \frac{\ln(1 + 2\lambda B')}{2\lambda B'}, \qquad B' = \frac{\dot{m}_c + \dot{m}_g}{\rho_e u_e C_H}, \tag{3.16}$$

with $\lambda \approx 0.5$ for laminar and $\lambda \approx 0.4$ for turbulent boundary layers. Note that $\phi \to 1$ as $B' \to 0$, so eq. (3.16) degrades gracefully to the non-ablating limit; as with eq. (4.7) below, the expression should be evaluated by its series expansion for small B' to avoid cancellation.

Remark 3.8. Equation (3.15) presumes a surface thermochemistry table supplying $\dot{s}$ as a function of (T_w, p_w, B'), generated offline by equilibrium minimization. PASSES treats this table as an input. It is the one element of the thermal model that is not evaluated from closed form at runtime, and it is interpolated — a departure from the C^∞ claim made for the rest of the framework, which we note rather than obscure. Table interpolation is performed with a C^2 tensor-product spline so that the discontinuity is in the third derivative rather than the first.

3.5 The global state vector and Method of Lines

The bulk rigid-body kinematics are coupled to the spectral grids in a single state vector. Six-degree-of-freedom rotational states are retained because the coupling between structural bending slope $\partial w/\partial x$ and inertial measurement unit orientation is a first-order effect on the navigation solution and cannot be recovered from translational states alone. The complete state is

$$\mathbf{X}_{\text{global}} = \begin{bmatrix} \mathbf{r}_E \\ \mathbf{v}_E \\ \mathbf{q}_{E2B} \\ \boldsymbol{\omega}_B \\ m_{\text{bulk}} \\ \hat{\mathbf{w}} \\ \dot{\hat{\mathbf{w}}} \\ \mathbf{H} \end{bmatrix} \in \mathbb{R}^n, \qquad n = 3 + 3 + 4 + 3 + 1 + 2(N-3) + N_T, \tag{3.17}$$

where $\hat{\mathbf{w}}$ is the null-space-reduced structural coordinate of eq. (3.6), and $\mathbf{H} \in \mathbb{R}^{N_T}$ is the vector of nodal in-depth enthalpies on the thermal grid, from which T is recovered by inverting the caloric equation of state.

Remark 3.9 ($\hat{\mathbf{w}}$ is the *elastic* deformation)**.** The free-free spectrum of eq. (3.6) contains two zero-frequency modes — rigid translation and rotation — and those are already carried by $\mathbf{r}_E$, $\mathbf{v}_E$, $\mathbf{q}_{E2B}$ and $\boldsymbol{\omega}_B$. The structural block of eq. (3.17) must therefore span the elastic modes only. Retaining the rigid modes in $\hat{\mathbf{w}}$ double-counts rigid motion and, because those modes have no restoring term, leaves a modal coordinate that integrates any steady load without bound; in a coupled entry simulation the coordinate diverges rather than merely being redundant. The requirement is easy to state and easy to violate, because each block is individually correct: the error appears only on assembly.

Enthalpy rather than temperature is carried as the state variable because it remains continuous across the pyrolysis zone where c_p varies sharply.

3.5.1 Quaternion norm drift

The attitude quaternion evolves by

$$\dot{\mathbf{q}}_{E2B} = \tfrac{1}{2}\,\boldsymbol{\Omega}(\boldsymbol{\omega}_B)\,\mathbf{q}_{E2B}, \qquad \boldsymbol{\Omega}(\boldsymbol{\omega}) = \begin{bmatrix} 0 & -\boldsymbol{\omega}^\top \\ \boldsymbol{\omega} & -[\boldsymbol{\omega}\times] \end{bmatrix}. \tag{3.18}$$

The unit-norm constraint $\|\mathbf{q}\| = 1$ is an invariant of the exact flow of eq. (3.18) but is not preserved by a general Runge–Kutta method, which drifts off the constraint manifold at the order of the local truncation error and accumulates over a long trajectory. Rather than renormalizing after each step — which is cheap but silently perturbs the embedded error estimate the adaptive controller relies on — PASSES applies Baumgarte stabilization inside the right-hand side,

$$\dot{\mathbf{q}}_{E2B} = \tfrac{1}{2}\boldsymbol{\Omega}(\boldsymbol{\omega}_B)\mathbf{q}_{E2B} + k_q\left(1 - \|\mathbf{q}_{E2B}\|^2\right)\mathbf{q}_{E2B}, \tag{3.19}$$

with $k_q > 0$ chosen so that $1/k_q$ is short relative to the trajectory but long relative to the step size. This renders the unit sphere attracting rather than merely invariant, and keeps the correction visible to the error controller.

3.6 Temporal integration and the explicit stability constraint

The semi-discrete system is advanced by an embedded Runge–Kutta–Fehlberg (RKF45) pair with adaptive step control. This choice interacts badly with the spectral structural operator, and the interaction must be stated rather than assumed away.

Proposition 3.10 (Explicit step-size limit)**.** *Let $\hat{\mathbf{K}}, \hat{\mathbf{M}}$ be as in eq.* (3.6) *with $\hat{\mathbf{M}} \succ 0$, and let $\lambda_{\max}$ be the largest generalized eigenvalue of $(\hat{\mathbf{K}}, \hat{\mathbf{M}})$. Written in first-order form, the undamped structural block has purely imaginary eigenvalues $\pm i\,\omega_{\max}$ with $\omega_{\max} = \sqrt{\lambda_{\max}}$. Any explicit Runge–Kutta method with a bounded stability region requires*

$$\Delta t \;\leq\; \frac{C_{RK}}{\omega_{\max}}, \tag{3.20}$$

where C_{RK} is the imaginary-axis stability limit of the method ($C_{RK} \approx 3.0$ for the fifth-order RKF member). Since $\lambda_{\max}$ inherits the $\mathcal{O}(N^8)$ growth of the fourth-derivative operator, $\omega_{\max} = \mathcal{O}(N^4)$ and hence $\Delta t = \mathcal{O}(N^{-4})$.

Proof. Immediate from the generalized eigendecomposition of the second-order system and the definition of the linear stability region; the growth rate of $\lambda_{\max}$ follows from that of the collocation representation of $\mathrm{d}^4/\mathrm{d}x^4$. □

Remark 3.11 (Consequence)**.** Proposition 3.10 is severe. At $N = 32$, $N^4 \approx 10^6$: an adaptive explicit integrator will select steps governed by structural modes far above the bandwidth of any physically meaningful excitation, and the trajectory cost becomes dominated by resolving modes that carry no energy. The framework does not escape this by virtue of being continuous, and any claim of an unbroken single-integrator trajectory must confront it.

Two mitigations are available, and PASSES is designed to accept either.

Modal truncation. Solve the generalized eigenproblem for $(\hat{\mathbf{K}}, \hat{\mathbf{M}})$ once per stage configuration and retain the lowest n_m modes, with n_m chosen so that ω_{n_m} exceeds the highest excitation frequency of interest by a stated margin. This caps $\omega_{\max}$ at a physically motivated value and restores a workable Δt. It is the default. The cost is a re-solve at each configuration change, and the approximation error is quantified by the retained-mode mass participation ratio.

IMEX splitting. Treat the linear structural block implicitly and the aerodynamic, thermal, and rigid-body couplings explicitly. Since $\hat{\mathbf{K}}$ is constant between configuration changes, its factorization is reused across all steps and across all Monte Carlo replicates, which preserves the batching argument of section 5. This removes the constraint of eq. (3.20) entirely at the cost of one triangular solve per stage.

Remark 3.12. The thermal block eq. (3.13) is a parabolic operator and carries its own explicit constraint $\Delta t = \mathcal{O}(\Delta\eta^2)$, which for the thermal grid resolutions of interest is typically less restrictive than eq. (3.20) but is not negligible. The same IMEX treatment applies.

Establishing which mitigation is preferable, and at what n_m, is verification task V3 in section 7.

Measured on the structural block alone, the explicit step obeys eq. (3.20): the achieved Δt scales as $N^{-3.6}$ over $N \in [12, 32]$ and sits at an $\mathcal{O}(1)$ multiple of the stability bound, confirming that steps are stability-limited rather than accuracy-limited. On the fully coupled system of eq. (3.17), a *fully implicit* method removes the constraint outright: with BDF the cost is flat — about 2×10^3 right-hand-side evaluations for a 110 s entry arc — across a nineteen-fold range of $\omega_{\max}$.

Remark 3.13 (Adaptive is not a substitute for implicit)**.** Two practical findings accompany that result and are worth recording, because both cost orders of magnitude in work and neither is visible in the formulation. First, an automatically stiffness-*switching* integrator is not equivalent to an implicit one: LSODA, applied to the coupled system, fails to detect the stiffness and requires 10^5–10^6 evaluations for an arc that BDF completes in 10^3. Second, the global state of eq. (3.17) spans roughly nine orders of magnitude in units — ECI position in 10^6 m against elastic modal coordinates that a free-free structure keeps near zero, since an elastic mode is mass-orthogonal to rigid translation and a uniform load barely excites it. A single scalar absolute tolerance cannot serve both: set it for position and the modal block is unconstrained, set it for the modal block and the integrator chases rounding in the orbital state. Per-component tolerances are not a refinement here but a requirement, worth three orders of magnitude in cost.

4 Guidance, Navigation, and Control Architecture

4.1 Anomaly detection by Mahalanobis distance

Let the EKF innovation and its theoretical covariance be

$$\boldsymbol{\nu}_k = \mathbf{z}_k - \mathbf{h}(\hat{\mathbf{x}}_{k|k-1}), \qquad \mathbf{S}_k = \mathbf{H}_k \mathbf{P}_{k|k-1} \mathbf{H}_k^\top + \mathbf{R}_k. \tag{4.1}$$

Under the nominal hypothesis — correct model, correct noise statistics, valid linearization — the normalized innovation squared

$$d_k^2 = \boldsymbol{\nu}_k^\top \mathbf{S}_k^{-1} \boldsymbol{\nu}_k \tag{4.2}$$

is distributed as χ^2 with $m = \dim(\mathbf{z}_k)$ degrees of freedom [2]. An anomaly is declared when

$$d_k^2 > \chi^2_{m,\,1-p}, \tag{4.3}$$

with p the false-alarm probability, taken as $p = 10^{-3}$ so that nominal flight does not trigger adaptation. Equation (4.3) is the geometric detector referred to throughout: it is a statement about the position of the innovation relative to the covariance ellipsoid the filter believes it should occupy, and it requires no model of the separation event itself.

Remark 4.1. Equation (4.2) inherits the validity of the EKF linearization. During the most violent part of a separation transient the linearization is itself suspect, so d_k^2 should be read as a detector of *model inadequacy* generally, not specifically of separation. This is adequate for the present purpose — the response, process noise inflation, is the appropriate response to model inadequacy from any cause — but it means the detector cannot be used to *identify* the event.

4.2 Innovation-Based Adaptive Estimation

On detection, the filter inflates its process noise. The sample innovation covariance over a sliding window of length N_w is

$$\hat{\mathbf{C}}_k = \frac{1}{N_w} \sum_{j=k-N_w+1}^{k} \boldsymbol{\nu}_j \boldsymbol{\nu}_j^\top, \tag{4.4}$$

and the inflation factor is

$$\alpha_k = \min\left(\alpha_{\max},\ \max\left(1,\ \frac{\max\left(0,\ \operatorname{tr}(\hat{\mathbf{C}}_k - \mathbf{R}_k)\right)}{\operatorname{tr}\left(\mathbf{H}_k \mathbf{P}_{k|k-1} \mathbf{H}_k^\top\right) + \epsilon}\right)\right), \qquad \mathbf{Q}_k^* = \alpha_k \mathbf{Q}_{\text{nom}}, \tag{4.5}$$

with $\epsilon > 0$ a fixed regularizer and $\alpha_{\max}$ a finite cap.

The window length N_w is a genuine design parameter, not an implementation detail: it sets the trade between the variance of $\hat{\mathbf{C}}_k$ (which falls as N_w^{-1}) and the lag before the adaptation responds (which grows as N_w). We take N_w corresponding to 0.5–2 s of measurement history, short relative to the recovery time of the transient and long enough that $\hat{\mathbf{C}}_k$ is not dominated by a single sample.

Proposition 4.2 (Well-posedness of the inflation)**.** *For any $\mathbf{Q}_{nom} \succeq 0$ and any realization of the innovation sequence, the inflated process noise satisfies $\mathbf{Q}_{nom} \preceq \mathbf{Q}_k^* \preceq \alpha_{\max}\mathbf{Q}_{nom}$, and in particular $\mathbf{Q}_k^* \succeq 0$.*

Proof. By eq. (4.5), $\alpha_k \in [1, \alpha_{\max}]$: the denominator is strictly positive by the ϵ regularizer, the numerator is non-negative by the inner max, and the outer max and min bound the result. For $\alpha \geq 1$ and $\mathbf{Q}_{\text{nom}} \succeq 0$, $\alpha\mathbf{Q}_{\text{nom}} - \mathbf{Q}_{\text{nom}} = (\alpha - 1)\mathbf{Q}_{\text{nom}} \succeq 0$, and similarly for the upper bound. Positive semidefiniteness is preserved under multiplication by a non-negative scalar. □

Remark 4.3 (What the clamps actually do)**.** It is worth being precise, because this is easy to overstate. Positive semidefiniteness of $\mathbf{Q}_k^*$ is *not* a consequence of the inner $\max(0, \cdot)$: it follows trivially from scaling a PSD matrix by a non-negative scalar, as in Proposition 4.2. A clamp on the scalar trace $\operatorname{tr}(\hat{\mathbf{C}}_k - \mathbf{R}_k)$ implies nothing about the definiteness of the matrix $\hat{\mathbf{C}}_k - \mathbf{R}_k$, which may well be indefinite. The inner max serves a different and more mundane purpose: it prevents a sample-covariance deficit — which occurs routinely when N_w is small and the innovations happen to be quiet — from driving α_k negative before the outer max sees it.

The structural reason this adaptation is well behaved is that it estimates a single scalar rather than a full matrix. Entrywise estimators of $\mathbf{Q}$ in the style of [15] can and do return indefinite matrices under short windows, and require projection back onto the PSD cone. Scalar inflation is strictly weaker — it cannot redistribute uncertainty between states, only scale it — and that weakness is what buys unconditional well-posedness. The $\alpha_{\max}$ cap is likewise not decorative: without it, a persistent modeling error drives α_k upward without bound, and the filter asymptotically ignores its dynamic model entirely.

Characterizing recovery time and the sensitivity of the terminal solution to $(N_w, \alpha_{\max}, p)$ is validation task V5 in section 7. Measured on a constant-velocity plant with exactly known nominal statistics, the gate fires at 9.8×10^{-4} against a design $p = 10^{-3}$ over 5×10^5 nominal evaluations, and no replicate diverges in any of twenty-seven $(N_w, \alpha_{\max}, p)$ configurations — which is Proposition 4.2 made empirical. Recovery from an injected 30 m/s transient takes a median 1.30 s against 2.15 s for the same filter with $\alpha_{\max} = 1$, on identical measurement realizations.

4.3 Terminal guidance: Aerodynamically-Compensated APN

Let $R_{\text{LOS}} = \|\mathbf{r}_{\text{LOS}}\|$ be the range to the target, $V_c = -\dot{R}_{\text{LOS}} > 0$ the closing velocity, and $A_c = \dot{V}_c$ the closing acceleration, all supplied by the EKF. Under constant A_c, the range satisfies $R(t) = R_{\text{LOS}} - V_c t - \frac{1}{2}A_c t^2$, and the time-to-go is the smallest positive root,

$$t_{go} = \frac{-\hat{V}_c + \sqrt{\hat{V}_c^2 + 2\hat{A}_c R_{\text{LOS}}}}{\hat{A}_c}. \tag{4.6}$$

Equation (4.6) is algebraically correct and numerically unusable. As $\hat{A}_c \to 0$ — the near-constant-closing-rate condition that holds over most of terminal flight — both numerator and denominator vanish, and the numerator suffers catastrophic cancellation between two nearly equal quantities before being divided by a small number. PASSES evaluates the mathematically equivalent conjugate form

$$\boxed{t_{go} = \frac{2R_{\text{LOS}}}{\hat{V}_c + \sqrt{\hat{V}_c^2 + 2\hat{A}_c R_{\text{LOS}}}}} \tag{4.7}$$

Proposition 4.4 (Stability of the time-to-go root)**.** *For $\hat{V}_c > 0$ and discriminant $\mathcal{D} = \hat{V}_c^2 + 2\hat{A}_c R_{LOS} \geq 0$, eq. (4.7) and eq. (4.6) agree exactly in exact arithmetic; eq. (4.7) is continuous at $\hat{A}_c = 0$ with $t_{go} \to R_{LOS}/\hat{V}_c$, involves no cancellation between like-signed quantities, and its denominator is bounded below by $\hat{V}_c > 0$.*

Proof. Multiply numerator and denominator of eq. (4.6) by $\hat{V}_c + \sqrt{\mathcal{D}}$. The numerator becomes $\mathcal{D} - \hat{V}_c^2 = 2\hat{A}_c R_{\text{LOS}}$, and the factor $\hat{A}_c$ cancels against the denominator, giving eq. (4.7). Both summands in the denominator are non-negative and the first is strictly positive, so no cancellation occurs and the denominator is bounded below by $\hat{V}_c$. Continuity at $\hat{A}_c = 0$ follows by direct substitution. □

Remark 4.5 (Non-intercept guard)**.** When $\hat{A}_c < 0$ — decelerating closure, the normal condition under atmospheric drag — the discriminant $\mathcal{D}$ can become negative, meaning the constant-acceleration extrapolation predicts that the vehicle never reaches the target. This is a physically meaningful condition, not a numerical failure, and it must not be allowed to propagate as a NaN. PASSES clamps to the linear estimate,

$$t_{go} = \begin{cases} 2R_{\text{LOS}}/\big(\hat{V}_c + \sqrt{\mathcal{D}}\big), & \mathcal{D} \geq 0, \\ R_{\text{LOS}}/\hat{V}_c, & \mathcal{D} < 0, \end{cases}$$

and raises a guidance-feasibility flag. In batch execution the flag is recorded per replicate, since a systematic pattern of infeasibility across the ensemble indicates an energy-management problem upstream rather than a guidance problem.

The commanded lateral acceleration is formed orthogonally to the relative velocity vector, with navigation gain scaled by dynamic pressure and feed-forward gravity compensation,

$$\mathbf{a}_n = N'\left(\dot{\boldsymbol{\lambda}} \times \mathbf{v}_{\text{rel}}\right) + \frac{N'}{2}\mathbf{a}_T - \mathbf{g}_\perp, \tag{4.8}$$

where $\dot{\boldsymbol{\lambda}}$ is the line-of-sight rate, $\mathbf{a}_T$ the estimated target acceleration, and $\mathbf{g}_\perp$ the component of gravity normal to $\mathbf{v}_{\text{rel}}$. The gravity term is feed-forward rather than corrective: without it the guidance law expends line-of-sight rate correcting a bias it could have anticipated, which shows up as a systematic downrange offset in the dispersion footprint.

5 Computational Architecture

5.1 Reproducibility

The framework is containerized. Deployment is managed by a Git-tracked Docker Compose stack and a VS Code DevContainer specification, with the build context established on an `nvidia/cuda:12.x-devel` base image so that the compilation of the underlying linear algebra backends is deterministic. Container digests, not tags, are pinned, so that a rebuild at a later date reproduces the same toolchain.

5.2 Memory architecture and parallelization

The mathematical structure admits a specific and unusually clean parallelization, which is the substantive computational claim of this paper.

The canonical Chebyshev differentiation matrices $\mathbf{D}$ depend only on N. They are identical across every Monte Carlo replicate and constant for the entire trajectory, so they are staged once into shared memory per streaming multiprocessor and read by every thread in the block. The mass and stiffness operators $\mathbf{M}(t)$, $\mathbf{K}(t)$ vary with fuel depletion and thermal softening and are reassembled by parallel kernels from eq. (3.4), but their sparsity pattern is fixed, so no symbolic analysis is repeated.

The consequence for batch execution is the following. Because the coupled system is one ODE with a fixed dimension and fixed sparsity, a batch of N_{MC} replicates differing only in their sampled parameters and initial conditions is a rank-3 tensor operation: replicate index, state index, stage index. No replicate requires a remesh, and under the IMEX strategy of section 3.6 the factorization of $\hat{\mathbf{K}}$ is shared across the entire batch. This is what a moving-mesh formulation cannot offer, since there each replicate's mesh diverges from every other's after the first remesh event and the batch decoheres.

Remark 5.1. Replicates that adapt their step sizes independently will diverge in step count, causing warp divergence within a block. The mitigation is to integrate the batch on a common outer time grid with per-replicate substepping, at the cost of some wasted work on replicates that could have taken longer steps. The efficiency of this trade is an empirical question. Measured on the batched entry workload, throughput scales linearly in N_{MC} below device saturation (fitted log–log slope 1.00) and peaks at roughly 8.5×10^4 replicates/s, about $5\times$ the vectorized CPU batch and $380\times$ a per-replicate loop — the decohered execution model a moving-mesh formulation forces. *Theoretical* occupancy of the batched stage kernel is computed exactly from its register and shared-memory footprint against the device's per-SM limits. *Achieved* occupancy and warp divergence are hardware counters requiring a profiler, and remain unmeasured: on the present host the driver restricts performance counters to administrators. The instrumentation is in place and the measurement is a permissions question, not a code one.

6 Stochastic Characterization of Terminal Dispersion

Given N_{MC} terminal impact points $\{(x_i, y_i)\}_{i=1}^{N_{\mathrm{MC}}}$ in a local tangent plane, let $\hat{\boldsymbol{\mu}}$ and $\boldsymbol{\Sigma}$ be the sample mean and sample covariance, and let $\boldsymbol{\Sigma} = \mathbf{U}\boldsymbol{\Lambda}\mathbf{U}^{\top}$ be its eigendecomposition with $\lambda_1 \geq \lambda_2 > 0$ and $\sigma_i = \sqrt{\lambda_i}$. The eigenvectors give the principal axes of the dispersion ellipse and the eigenvalues its squared semi-axis scale.

6.1 Containment radius

Under a bivariate normal model, the locus of constant Mahalanobis radius c encloses probability $1 - \exp(-c^2/2)$. The 95% containment ellipse therefore corresponds to $c^2 = \chi^2_{2,\,0.95} = 5.991$, giving semi-axes

$$a_{95} = 2.4477\,\sigma_1, \qquad b_{95} = 2.4477\,\sigma_2, \tag{6.1}$$

and, when a single scalar radius is required, $R_{95} = a_{95}$ as the conservative circumscribing choice. Reporting R_{95} as a scalar discards the anisotropy, which for a lifting reentry is typically substantial; the eigenvalue pair should be reported alongside it.

6.2 Circular Error Probable

CEP is the radius of the circle containing 50% of impacts. For the isotropic case $\sigma_1 = \sigma_2 = \sigma$ it is exactly

$$\mathrm{CEP} = \sigma\sqrt{2\ln 2} \approx 1.1774\,\sigma. \tag{6.2}$$

For the anisotropic case no closed form exists, and PASSES uses the standard linear approximation

$$\mathrm{CEP} \approx 0.5887\,(\sigma_1 + \sigma_2), \tag{6.3}$$

which agrees with the exact value to better than 1% provided

$$0.25 \;\leq\; \sigma_2/\sigma_1 \;\leq\; 1. \tag{6.4}$$

Remark 6.1 (Validity)**.** Equation (6.4) is a real restriction and is routinely violated by lifting reentry footprints, which are strongly elongated downrange. PASSES evaluates the ratio σ_2/σ_1 for every batch and, when eq. (6.4) fails, falls back to the direct order statistic — the median of $\|\mathbf{p}_i - \hat{\boldsymbol{\mu}}\|$ over the sample — rather than reporting a CEP the approximation does not support. Reporting eq. (6.3) outside its validity range is a common and quiet source of error in dispersion analyses.

6.3 Sampling error

$\boldsymbol{\Sigma}$ is itself an estimate. For a bivariate normal sample, the relative standard error of each σ_i is approximately $1/\sqrt{2N_{\mathrm{MC}}}$, which propagates directly to CEP and R_{95}. At $N_{\mathrm{MC}} = 10^4$ this is 0.7%; at $N_{\mathrm{MC}} = 10^3$ it is 2.2%. Any dispersion figure reported without an accompanying sample size, or quoted to more significant figures than this bound supports, is not meaningful. All reported dispersion metrics carry bootstrap confidence intervals.

Remark 6.2. The bivariate normal assumption underlying eq. (6.1) and eq. (6.3) should be tested, not assumed. Nonlinear guidance and hard constraint boundaries produce skewed and occasionally bimodal footprints, for which an ellipse is a poor summary. PASSES applies a Henze–Zirkler test for bivariate normality to each batch and reports the result alongside the dispersion metrics.

7 Verification and Validation Plan

This section states what is measured, against what reference, and what constitutes failure. It was written before any implementation existed, so the plan is falsifiable in advance rather than reported selectively afterward. The tasks have since been executed against a reference implementation; measured values are quoted inline in the relevant sections above, and full numerical detail is deferred to the companion experimental report.

ID	Target	Method and reference	Failure criterion
V1	Structural operator	$\kappa(\hat{\mathbf{K}})$ versus N for uniform and stepped EI profiles; free-free natural frequencies against the analytic uniform-beam solution	relative frequency error $> 10^{-6}$ at $N = 32$ for the uniform case
V2	Slosh regularization	Total force and first moment transferred versus σ, N, and station x_s; Proposition 3.4	force error above machine precision; moment error not $\mathcal{O}(\sigma^2)$ in the interior
V3	Time integration	Achieved Δt and wall-clock for explicit, modally truncated, and IMEX strategies; Proposition 3.10	explicit Δt not scaling as N^{-4}
V4	Ablation	Method of manufactured solutions on eq. (3.13)–eq. (3.14); in-depth temperature and recession against a FIAT [4] reference case	recession disagreement $> 5\%$ on the reference case
V5	Adaptive filter	Recovery time after an injected separation transient; sensitivity to $(N_w, \alpha_{\max}, p)$; false-alarm rate against the design $p = 10^{-3}$	divergence on any replicate; measured false-alarm rate above $2p$
V6	Guidance	eq. (4.7) against eq. (4.6) in double and single precision as $\hat{A}_c \to 0$; Proposition 4.4	eq. (4.7) losing more than 1 significant digit where eq. (4.6) loses many
V7	Dispersion	Convergence of CEP and R_{95} with N_{MC}; bootstrap intervals; Henze–Zirkler normality	CEP not converging at the $1/\sqrt{2N_{\mathrm{MC}}}$ rate
V8	Throughput	Replicates per second versus N_{MC} and N; achieved occupancy; CPU baseline comparison	sublinear scaling in N_{MC} below device saturation

Status. V1, V2, V3, V5, V6, V7 and V8 executed and passing against the criteria as stated, with the exception noted for V8's occupancy counters. V4 is partially complete: the method-of-manufactured-solutions leg passes with spectral convergence to the time-integration floor, but the FIAT comparison requires reference-case data not available to this work.

8 Limitations

We state the boundaries of the formulation explicitly.

1. **Scope of the measurements.** The verification tasks of section 7 have been executed against a reference implementation and the results are quoted inline, but they are verification — internal consistency, convergence, and conservation — not validation against flight or ground-test data. Two comparisons requiring external reference data (the FIAT ablation case of V4, and published Fay–Riddell conditions in the companion paper) remain open, and no synthetic stand-in is offered for either. For V4 the situation has since been narrowed rather

than closed: FIAT is US-government-controlled software, so we implemented its published formulation independently from [4] and [16] — conservative finite volume, backward Euler, analytic Newton Jacobian — giving two structurally different solvers for the same physics. They agree, and Remark 3.6 explains why that is worth stating and why it is not the stated criterion.

2. **Numerical practice is part of the method.** Three results only appeared on assembly and none is implied by the formulation: the structural coordinate must exclude rigid modes (Remark 3.9); the coupled system requires a fully implicit integrator rather than an adaptive stiffness-switching one, and per-component absolute tolerances rather than a scalar (Remark 3.13); and boundary conditions that a manufactured-solution test supplies explicitly must be imposed by some other means in flight, here by constraint differentiation with Baumgarte stabilization. A reader implementing this formulation from the equations alone would meet all three.

3. **Euler–Bernoulli kinematics.** Assumption 3.1 excludes transverse shear and rotary inertia, which are not negligible for thick sections or high-frequency modes. The companion paper [13] addresses this with a Mindlin–Reissner formulation; this paper does not.

4. **Linear structural response.** Equation (3.1) is linear in w. Geometric nonlinearity under large deflection, and material nonlinearity under thermal softening beyond the linear-elastic range, are outside scope.

5. **One-dimensional thermal response.** Equation (3.13) resolves depth only. Lateral conduction, which matters near sharp leading edges and material joints, is neglected.

6. **Interpolated surface thermochemistry.** As noted in section 3.4, the B' table is interpolated. The framework is not C^∞ through this path, and the claim of global smoothness should be understood to exclude it.

7. **Aerodynamic closure.** Loads derive from engineering-level correlations, not from a flow solver. In regimes where those correlations are poor — separated flow, shock-boundary-layer interaction, base flow — the structural and thermal responses inherit that error regardless of how accurately they are integrated. Numerical smoothness is not physical accuracy, and the two should not be conflated.

8. **Linearization-dependent detection.** As noted in section 4.1, d_k^2 is only as valid as the EKF linearization it is computed from.

9. **Gaussian input model.** The dispersion analysis of section 6 presumes the sampled input uncertainties are correctly specified. Terminal dispersion is generally more sensitive to the input distributions than to the fidelity of the propagator, and no amount of numerical rigor downstream compensates for a misspecified input covariance.

9 Conclusion

This paper presented PASSES, a flight simulation framework in which structural, thermal, and rigid-body domains are discretized once on fixed computational grids and advanced as a single system of ordinary differential equations. Three formulations make the fixed grid possible: null-space projection for free-free spectral boundary conditions, a Landau transformation that renders the ablation front stationary in computational coordinates, and a quadrature-normalized kernel that transfers localized slosh forces exactly.

We have been explicit about what the formulation costs. Proposition 3.10 establishes that the spectral structural operator imposes an $\mathcal{O}(N^{-4})$ explicit step-size limit that continuity does not remove, and we give two mitigations rather than assume the problem away. Remark 4.3 corrects a tempting but incorrect reading of why the adaptive filter's inflation factor is well posed. Proposition 4.4 replaces the standard time-to-go root with an algebraically equivalent form that does not lose precision in the regime where terminal guidance actually operates. Remark 6.1 bounds the validity of the CEP approximation and specifies the fallback outside it.

The framework's central computational claim — that a fixed sparsity pattern with no mid-flight remeshing renders a Monte Carlo batch a single batched tensor operation — follows from the structure of the formulation. Whether it delivers the throughput that motivates it is an empirical question, and section 7 states in advance how it will be answered.

Code and Data Availability

The manuscript sources, bibliographies, and citation audit for this work are available at `https://github.com/ethank5149/Parameterized-AeroStructural-State-Estimation-Sandbox`. The PASSES implementation, container specifications, and the scripts generating every figure in the companion experimental manuscript will be released under the MIT license in the same repository. **[PENDING: archive a tagged release with a Zenodo DOI at submission]**

Acknowledgments

[PENDING: acknowledgments, if any]

References

[1] H. Norman Abramson, ed. *The Dynamic Behavior of Liquids in Moving Containers, with Applications to Space Vehicle Technology*. NASA SP-106. Washington, DC: National Aeronautics and Space Administration, 1966.

[2] Yaakov Bar-Shalom, X.-Rong Li, and Thiagalingam Kirubarajan. *Estimation with Applications to Tracking and Navigation: Theory, Algorithms and Software*. New York: John Wiley & Sons, 2001. DOI: 10.1002/0471221279.

[3] David A. Benson et al. "Direct trajectory optimization and costate estimation via an orthogonal collocation method". In: *Journal of Guidance, Control, and Dynamics* 29.6 (2006), pp. 1435–1440. DOI: 10.2514/1.20478.

[4] Y.-K. Chen and Frank S. Milos. "Ablation and thermal response program for spacecraft heatshield analysis". In: *Journal of Spacecraft and Rockets* 36.3 (1999), pp. 475–483. DOI: 10.2514/2.3469.

[5] John Crank. *Free and Moving Boundary Problems*. Oxford: Oxford University Press, 1984.

[6] John A. Dec and Robert D. Braun. "An approximate ablative thermal protection system sizing tool for entry system design". In: *44th AIAA Aerospace Sciences Meeting and Exhibit*. AIAA Paper 2006-780. Reno, NV, 2006. DOI: 10.2514/6.2006-780.

[7] Franklin T. Dodge. *The New "Dynamic Behavior of Liquids in Moving Containers"*. San Antonio, TX: Southwest Research Institute, 2000.

[8] Tobin A. Driscoll and Nicholas Hale. "Rectangular spectral collocation". In: *IMA Journal of Numerical Analysis* 36.1 (2016), pp. 108–132. DOI: 10.1093/imanum/dru062.

[9] Gamal Elnagar, Mohammad A. Kazemi, and Mohsen Razzaghi. "The pseudospectral Legendre method for discretizing optimal control problems". In: *IEEE Transactions on Automatic Control* 40.10 (1995), pp. 1793–1796. DOI: 10.1109/9.467672.

[10] Chris Hide, Terry Moore, and Martin Smith. "Adaptive Kalman filtering for low-cost INS/GPS". In: *The Journal of Navigation* 56.1 (2003), pp. 143–152. DOI: 10.1017/S0373463302002151.

[11] C. P. Hsu, C. F. Hung, and J. Y. Liao. "A Chebyshev spectral method with null space approach for boundary-value problems of Euler–Bernoulli beam". In: *Shock and Vibration* 2018 (2018), p. 2487697. DOI: 10.1155/2018/2487697.

[12] Geoffrey T. Huntington and Anil V. Rao. "Optimal reconfiguration of spacecraft formations using the Gauss pseudospectral method". In: *Journal of Guidance, Control, and Dynamics* 31.3 (2008), pp. 689–698. DOI: 10.2514/1.31083.

[13] Ethan Knox. *PASSES-HGV: Expanding the Continuous Spectral Framework to 6-DOF Hypersonic Glide and Fractional Orbital Trajectories*. Companion manuscript. TODO: insert arXiv identifier once posted. 2026.

[14] Raman K. Mehra. "Approaches to adaptive filtering". In: *IEEE Transactions on Automatic Control* 17.5 (1972), pp. 693–698. DOI: 10.1109/TAC.1972.1100100.

[15] Raman K. Mehra. "On the identification of variances and adaptive Kalman filtering". In: *IEEE Transactions on Automatic Control* 15.2 (1970), pp. 175–184. DOI: 10.1109/TAC.1970.1099422.

[16] Frank S. Milos, Yih-Kanq Chen, and Thomas H. Squire. "Updated ablation and thermal response program for spacecraft heatshield analysis". In: *17th Thermal and Fluids Analysis Workshop*. TFAWS06-1008. NASA Ames Research Center. University of Maryland, 2006.

[17] A. H. Mohamed and K. P. Schwarz. "Adaptive Kalman filtering for INS/GPS". In: *Journal of Geodesy* 73.4 (1999), pp. 193–203. DOI: 10.1007/s001900050236.

[18] Joshua D. Monk et al. "MEDLI2 material response model development and validation". In: *AIAA Aviation Forum*. NASA Ames Research Center; transcribed from the archived PDF in reference/. 2021.

[19] Carl B. Moyer and Roald A. Rindal. *An Analysis of the Coupled Chemically Reacting Boundary Layer and Charring Ablator. Part II: Finite Difference Solution for the In-Depth Response of Charring Materials Considering Surface Chemical and Energy Balances.* NASA Contractor Report CR-1061. Aerotherm Corporation, Palo Alto, CA. Washington, DC: National Aeronautics and Space Administration, 1968.

[20] Sheehan Olver and Alex Townsend. "A fast and well-conditioned spectral method". In: *SIAM Review* 55.3 (2013), pp. 462–489. DOI: 10.1137/120865458.

[21] Jing Pei. "Analytical investigation of propellant slosh stability boundary on a space vehicle". In: *Journal of Spacecraft and Rockets* 58.5 (2021), pp. 1514–1521. DOI: 10.2514/1.A35024.

[22] Wei Shyy et al. *Computational Fluid Dynamics with Moving Boundaries.* Series in Computational and Physical Processes in Mechanics and Thermal Sciences. Washington, DC: Taylor & Francis, 1996.

[23] Francisco Torres-Herrador et al. "A high heating rate pyrolysis model for the Phenolic Impregnated Carbon Ablator (PICA) based on mass spectroscopy experiments". In: *Journal of Analytical and Applied Pyrolysis* 141 (2019), p. 104625.

[24] Francisco Torres-Herrador et al. "Competitive kinetic model for the pyrolysis of the Phenolic Impregnated Carbon Ablator". In: *Aerospace Science and Technology* 100 (2020), p. 105784.

[25] Lloyd N. Trefethen. *Spectral Methods in MATLAB.* Software, Environments, and Tools. Philadelphia, PA: Society for Industrial and Applied Mathematics, 2000. DOI: 10.1137/1.9780898719598.

[26] Paul Zarchan. *Tactical and Strategic Missile Guidance.* 6th ed. Vol. 239. Progress in Astronautics and Aeronautics. Reston, VA: American Institute of Aeronautics and Astronautics, 2012. ISBN: 978-1-60086-894-8. DOI: 10.2514/4.868948.

Appendix A Chebyshev Differentiation Matrices

On the Chebyshev–Gauss–Lobatto nodes $\xi_j = \cos(j\pi/N)$, $j = 0, \dots, N$, the entries of the first-order differentiation matrix are

$$D_{ij} = \frac{c_i}{c_j}\frac{(-1)^{i+j}}{\xi_i - \xi_j}\ (i \neq j), \qquad D_{jj} = -\frac{\xi_j}{2(1-\xi_j^2)}\ (1 \leq j \leq N-1), \tag{A.1}$$

with $D_{00} = (2N^2+1)/6$, $D_{NN} = -(2N^2+1)/6$, and $c_0 = c_N = 2$, $c_j = 1$ otherwise [25].

Higher derivatives are formed as matrix powers, $\mathbf{D}^{(k)} = \mathbf{D}^k$. This is convenient but not benign: rounding error in $\mathbf{D}$ is amplified by roughly $\kappa(\mathbf{D})$ at each multiplication, and by $k = 4$ the accumulated error is appreciable at large N. Two standard remedies apply. The diagonal entries should be computed by the negative-sum trick, $D_{jj} = -\sum_{i\neq j} D_{ji}$, which enforces the exact annihilation of constants and materially improves accuracy. Where N is large enough that this is insufficient, the fourth-derivative matrix should be constructed directly from the recurrence rather than by repeated multiplication. The choice of N at which the direct construction becomes necessary is part of verification task V1.

Appendix B EKF Jacobians and Linearization

The discrete state transition matrix is obtained by linearizing the continuous plant $\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}, \mathbf{u})$ about the current estimate and propagating over the measurement interval,

$$\mathbf{F}_k = \left.\frac{\partial \mathbf{f}}{\partial \mathbf{x}}\right|_{\hat{\mathbf{x}}_{k|k},\, \mathbf{u}_k}, \qquad \mathbf{\Phi}_k = \exp\left(\mathbf{F}_k \Delta t\right) \approx \mathbf{I} + \mathbf{F}_k \Delta t + \tfrac{1}{2}\mathbf{F}_k^2 \Delta t^2, \tag{B.1}$$

with the observation Jacobian $\mathbf{H}_k = \partial \mathbf{h}/\partial \mathbf{x}$ evaluated at the prior. The second-order truncation in eq. (B.1) is retained rather than the first-order form because $\mathbf{F}_k \Delta t$ is not uniformly small during high-rate maneuvers.

The discrete process noise is obtained by the Van Loan method rather than by the common approximation $\mathbf{Q}_k \approx \mathbf{Q}\Delta t$, which is inconsistent with the second-order $\mathbf{\Phi}_k$ above and biases the covariance low at large Δt.

Remark B.1. The structural states enter $\mathbf{H}_k$ through the bending slope at the IMU station: a strapdown unit rigidly mounted to a flexing airframe measures the local rotation rate, which includes $\partial^2 w/\partial x \partial t$ evaluated at the mounting station. Omitting this term is the standard practice of treating the airframe as rigid for navigation purposes, and it is precisely the coupling that motivates carrying the 6-DOF rotational states in eq. (3.17).